THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue ____________________
Seat Number __________
Student Number |__|__|__|__|__|__|__|__|
Family Name ____________________
First Name ____________________

School of Mathematics & Physics
Semester 1 Examinations, 2025
STAT1201 Analysis of Scientific Data

This paper is for Gatton Campus and St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- No written or printed material permitted
- Casio FX82 series or UQ approved and labelled calculator only
- During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are 100 marks available on this exam from 5 questions.
Write your answers in the spaces provided in pages 2–15 of this examination paper. Show your working and state conclusions, where appropriate.
Pages 16–18 give formulas and probability values. Those pages will not be marked.

For Examiner Use Only

Question	Mark
Q1a	
Q1b	
Q1c	
Q2a	
Q2b	
Q2c	
Q2d	
Q3a	
Q3b	
Q3c	
Q3d	
Q3e	
Q4a	
Q4b	
Q4c	
Q4d	
Q4e	
Q5a	
Q5b	
Q5c	

Total __________

Question 1 [13 marks]

Neural spike counts are used by researchers to understand neural activity and information processing in the brain. Researchers studying the primary visual cortex of cats have observed that neural spike counts during 100 millisecond (ms) sampling windows can be well approximated by the following distribution.

x	0	1	2	3
$P(X = x)$	0.5	0.35	0.12	0.03

a) What are the expected value and standard deviation of X? *[5 marks]*

$E(X) = 0 \times 0\cdot5 + 1 \times 0\cdot35 + 2 \times 0\cdot12 + 3 \times 0\cdot03 = 0\cdot68$

$Var(X) = (0-0\cdot68)^2 \times 0\cdot5 + (1-0\cdot68)^2 \times 0\cdot35 + (2-0\cdot68)^2 \times 0\cdot12$
$+ (3-0\cdot68)^2 \times 0\cdot03 = 0\cdot6376$

$sd(X) = \sqrt{0\cdot6376} = 0\cdot798$

b) Suppose a researcher observes 4 independent 100 ms intervals. What is the probability that no more than one neural spike is observed in each interval? *[4 marks]*

Probability of no more than one in an interval

$P(X \leq 1) = P(X = 0) + P(X = 1) = 0\cdot85$

Probability that no more than one in each interval

$P(X_1 \leq 1, \ldots, X_4 \leq 1) = 0\cdot85^4 = 0\cdot522$

(by independence)

c) Suppose a researcher observes 4 independent 100 ms intervals. What is the expected value and standard deviation of the total number of neural spikes observed over the 4 intervals? *[4 marks]*

X_i – neural spike count for interval i

$T = X_1 + \cdots + X_4$

$E(T) = E(X_1) + \cdots + E(X_4) = 4 \times 0.68 = 2.72$

$Var(T) = Var(X_1) + \cdots + Var(X_4) = 4 \times 0.6376 = 2.55$

(by independence)

$sd(T) = \sqrt{2.55} = 1.60$

Question 2 [28 marks]

The effect of a liberal red-cell transfusion strategy as compared with a restrictive strategy in patients during the critical care period after an aneurysmal subarachnoid haemorrhage, a type of stroke, is unclear. To examine this issue, a study recruited 72 patients and randomly allocated them with equal probability to either a liberal or restrictive transfusion strategy.

a) The primary outcome was whether the patient suffered an unfavourable neurologic outcome after 12 months. The following table summarises the number of patients who experienced an unfavourable neurological outcome for each transfusion strategy.

	Unfavourable Outcome		
Strategy	Yes	No	
Liberal	11	25	36
Restrictive	16	20	36
	27	45	72

Carry out a χ^2-test to determine whether there is any evidence of a difference in the rate of unfavourable neurological outcomes for the two transfusion strategies. Remember to provide an appropriate conclusion.

[8 marks]

Expected counts

Liberal + Yes $= \frac{27 \times 36}{72} = 13.5$ Liberal + No $= \frac{45 \times 36}{72} = 22.5$

Restrictive + Yes $= \frac{27 \times 36}{72} = 13.5$ Restrictive + No $= \frac{45 \times 36}{72} = 22.5$

$$X^2 = \sum_i \frac{(o_i - e_i)^2}{e_i} = \frac{(11-13.5)^2}{13.5} + \frac{(16-13.5)^2}{13.5} + \frac{(25-22.5)^2}{22.5} + \frac{(20-22.5)^2}{22.5}$$

$= 1.4815$

degrees of freedom $= (r-1) \times (c-1) = 1$

P-value $= \mathbb{P}(\chi^2_1 \geq 1.4815) = 1 - 0.776 = 0.224$

There is inconclusive evidence against the null hypothesis. This suggests there is no difference in the rate of unfavourable outcomes.

[Extra space for Question 2a if needed]

b) A secondary outcome was the patient's functional independence after 12 months as assessed with the Functional Independence Measure (FIM; scores range from 18 to 126). The 36 patients allocated to the liberal strategy had an average FIM score of 92.4 with a standard deviation 54.6. The 36 patients allocated to the restrictive strategy had an average FIM score of 79.8 with a standard deviation 54.5.

Carry out an appropriate test to determine whether there is evidence of a difference in mean FIM scores after 12 months for the two transfusion strategies. Remember to state the hypotheses being tested and provide an appropriate conclusion. *[10 marks]*

Let μ_L and μ_R denote the mean FIM score of patients having the liberal & restrictive strategies, respectively.

$$H_0: \mu_L = \mu_R \quad \text{vs} \quad H_1: \mu_L \neq \mu_R$$

$$se(\bar{y}_L - \bar{y}_R) = \sqrt{\frac{s_L^2}{n_L} + \frac{s_R^2}{n_R}} = \sqrt{\frac{54.6^2}{36} + \frac{54.5^2}{36}} = 12.86$$

test statistic $t = \dfrac{(\bar{y}_L - \bar{y}_R) - 0}{se(\bar{y}_L - \bar{y}_R)} = \dfrac{(92.4 - 79.8) - 0}{12.86} = 0.98$

degrees of freedom $= \min(n_L - 1, n_R - 1) = 35$

P-value $= 2 \times P(T_{35} \geq 0.98) = 2 \times (1 - 0.833) = 0.334$

There is inconclusive evidence against the null hypothesis. This suggests there is no difference in mean FIM scores between the two treatment strategies.

c) In this study the recruited patients were randomly allocated to the two treatment groups. Why would it have been inappropriate to assign the 36 recruited patients recruited in the first month of the study to the liberal strategy and the 36 patients recruited in the second month to the restrictive strategy? How would that affect the conclusion?

[4 marks]

The treatment effect would be confounded with the time ordering of the treatments.

If we had observed a difference in mean FIM scores, we would not have been able to conclude the cause was the treatment strategy.

d) Construct a 95% confidence interval for the mean FIM score after 12 months for those who receive the liberal transfusion strategy. *[6 marks]*

$$t_{0.975;n-1} = t_{0.975;35} = 2.030$$

$$se(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{54.6}{\sqrt{36}} = 9.1$$

$$moe = 2.030 \times 9.1 = 18.47$$

The 95% CI for mean FIM score ~~to~~ with liberal strategy is

$$92.4 \pm \cancel{18.5}\ 18.47$$

Question 3 [22 marks]

Soil pH is one of the most important factors determining the chemical and biological properties of a soil. Soil pH is dependent on both the soil properties as well as the chemical composition of the equilibration solution used during pH measurement. For this reason, a method of converting soil pH values measured using different solutions is required.

To establish the relationship between soil pH values measured using H_2O and soil pH values measured with a $CaCl_2$ solution, the pH of 57 soil samples was measured using both. A linear regression model was then fitted to the data in R. Edited output from the regression summary is shown below.

```
Coefficients:
             Estimate Std. Error
(Intercept)   0.45061    0.19129
H2O           0.86327    0.02979
```

a) Write out the regression model that was fitted in R. *[2 marks]*

$$pH(CaCl_2) \sim N\left(0{\cdot}45 + 0{\cdot}86 \times pH(H2O),\ \sigma\right)$$

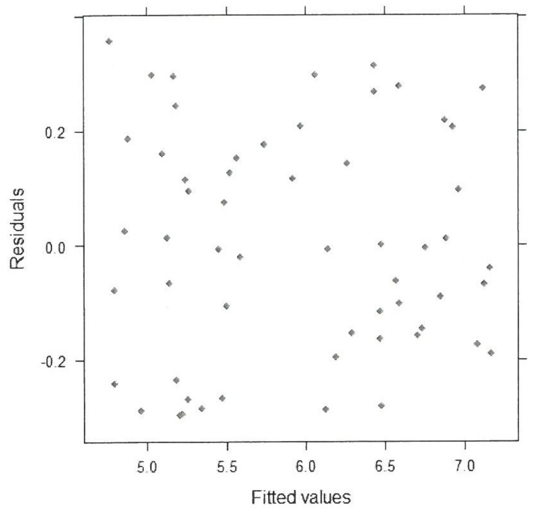
b) One soil sample had a pH measurement of 7.1 using the $CaCl_2$ solution and a pH measurement of 7.8 using H_2O. Determine the residual for this observation. *[2 marks]*

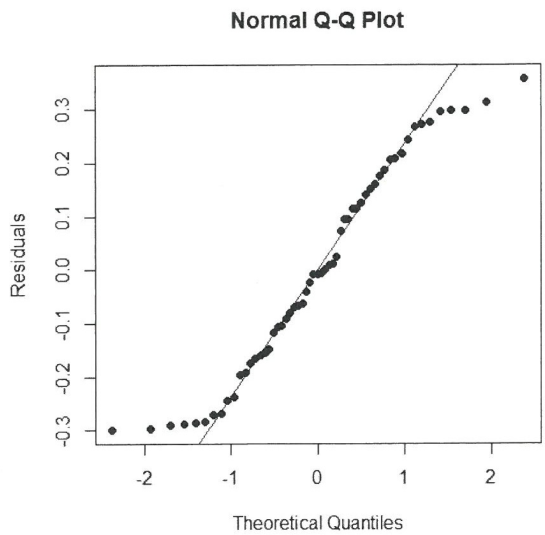
fitted value $= 0{\cdot}4506 + 0{\cdot}8633 \times 7{\cdot}8$

$= 7{\cdot}184$

residual $= 7{\cdot}1 - 7{\cdot}184 = -0{\cdot}084$

c) The following plots were obtained for the regression model in R:





Briefly explain what these plots are telling you about the model. *[6 marks]*

The plot of residuals vs fitted values shows

- there is no trends, indicating the relationship is linear
- the spread of the residuals is constant, indicating the constant variance /std deviation assumption is valid.

The quantile plot suggests the residuals are not normally distributed since it deviates from the straight line.

d) Regardless of your observations in Part (c), does this model give any evidence that the intercept term in the true model is zero.

[6 marks]

We want to test

$H_0 : \beta_0 = 0$ vs $H_1 : \beta_0 \neq 0$

β_0 – intercept in true regression model.

test statistic $t = \frac{0.4506 - 0}{0.1913} = 2.36$

$P\text{-value} = 2 \times P(T_{55} \geqslant 2.36) = 2 \times (1 - 0.989) = 0.022$

There is moderate evidence suggesting the intercept in the true regression model is not zero

e) Regardless of your observations in Part (c), construct a 95% confidence interval for the slope in the true model. *[6 marks]*

$t_{0.975;55} = 2.004$

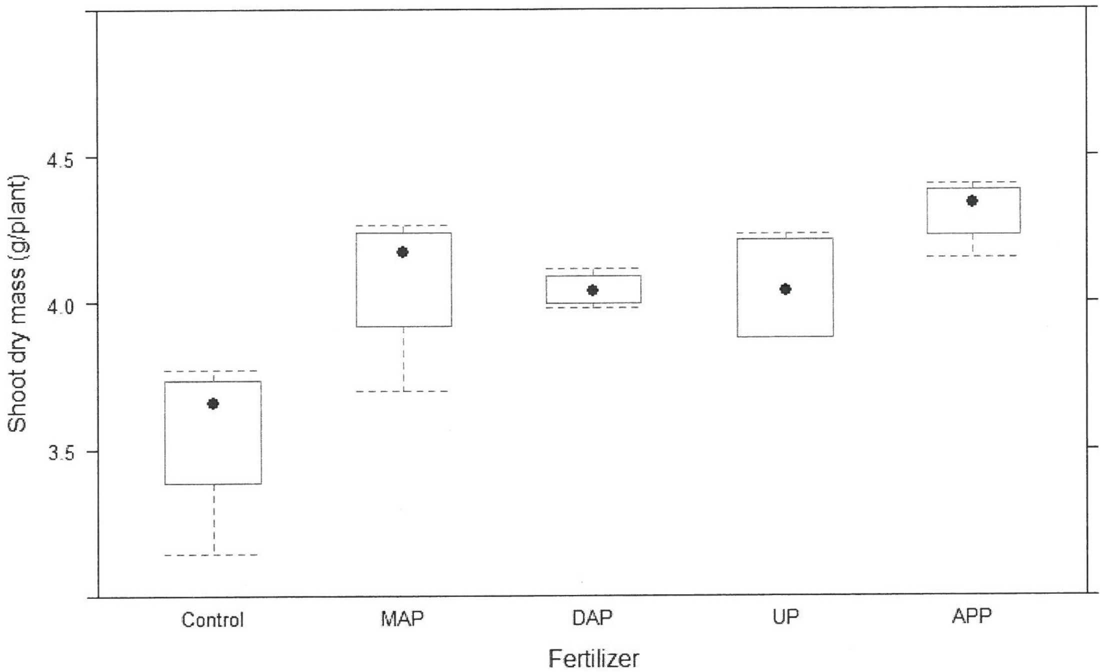
$se(\hat{\beta}_1) = 0.02979$

$moe = t_{0.975;55} \times se(\hat{\beta}_1) = 0.0597$

The 95% confidence interval for the slope is

0.863 ± 0.0597.

Question 4 [24 marks]
An agricultural experiment was conducted in 2018 to examine the effects of different phosphate fertilizers on maize growth in soil with high calcium carbonate content. In the experiment, the control (no added phosphate fertiliser) was compared with four different phosphate fertilizers applied broadcast. The fertilizers were monoammonium phosphate (MAP), diammonium phosphate (DAP), urea phosphate (UP) and ammonium polyphosphate (APP). These five treatments were applied to 20 plots in a completely randomized design with each treatment being applied to four plots. The shoot dry biomass of maize (g / plant) was measured 32 days after sowing. Side-by-side boxplots of the data are given below.



You decide to analyse the data from this experiment using ANOVA.

a) State the assumptions for one way ANOVA. *[3 marks]*

- Groups are independent
- residuals have a normal distribution
- variability of residuals does not depend on group.

For the remainder of Question 4 you may consider the assumptions for one-way ANOVA are satisfied.

b) Complete the ANOVA table by giving the appropriate degrees of freedom and calculating the mean sums of squares, F-statistic, and P-value. *[8 marks]*

Source	**df**	**SS**	**MS**	**F**	**P**
Fertilizer	4	1.1946	0.2986	7.54	0.0015
Residual	15	0.5936	0.03957		

$n = 20$ $\quad DFT = 20 - 1 = 19$ $\quad DFG = 5 - 1 = 4$ $\quad DFR = 19 - 4 = 15$

$$MSG = \frac{SSG}{DFG} = \frac{1.1946}{4} = 0.2986$$

$$MSR = \frac{SSR}{DFR} = \frac{0.5936}{15} = 0.03957$$

$$F = \frac{MSG}{MSR} = \frac{0.2986}{0.03957} = 7.54$$

$$\text{P-value} = P(F_{4,15} \geq 7.54) = 1 - 0.9985 = 0.0015$$

c) Briefly summarise the results from this analysis. *[3 marks]*

There is strong evidence that the mean growth rate of maize is different for at least one of the treatments (control or one of four fertilizers).

You decide to carry out post-hoc tests comparing the means using Tukey's Honestly Significant Difference in R. Output from your analysis is given below.

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

                     diff         lwr       upr     p adj
MAP-Control  0.519986146  0.08561841 0.9543539 0.0156763
DAP-Control  0.485097668  0.05072993 0.9194654 0.0252742
UP-Control   0.486709730  0.05234199 0.9210775 0.0247253
APP-Control  0.747820816  0.31345308 1.1821886 0.0007039
DAP-MAP     -0.034888478 -0.46925622 0.3994793 0.9990629
UP-MAP      -0.033276416 -0.46764415 0.4010913 0.9992219
APP-MAP      0.227834670 -0.20653307 0.6622024 0.5080223
UP-DAP       0.001612062 -0.43275568 0.4359798 1.0000000
APP-DAP      0.262723148 -0.17164459 0.6970909 0.3744114
APP-UP       0.261111086 -0.17325665 0.6954788 0.3801313
```

d) Assuming a 5% significance level, what is your conclusion from the pairwise comparisons analysis? *[6 marks]*

At the 5% level, there is evidence that the mean growth rate ~~is~~ under control is different to mean growth rate ~~of~~ under each fertiliser.

No evidence of a difference ~~betw~~ in mean growth rates between any of the fertilisers.

e) Why was it important to use a method like Tukey's Honestly Significant Difference when performing pairwise comparisons rather than just using standard two sample t-tests? *[4 marks]*

If we used standard two sample t-tests at the 5% level, the chance of at least one type I error from 10 tests will be greater than 5%. Tukey HSD ensures the chance of a type I error overall is still 5%, assuming no difference in means.

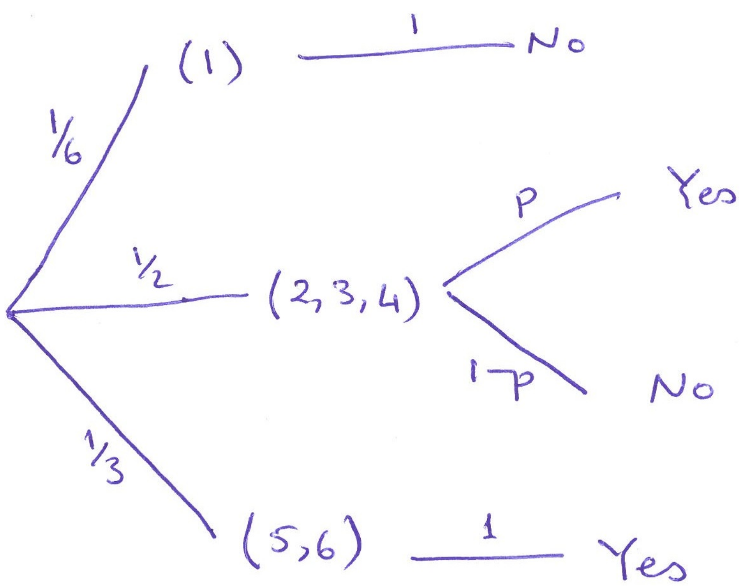
Question 5 [12 marks]

Measuring the prevalence of doping in recreational sport is difficult due to social desirability bias, that is survey respondents tend to answer in a way they believe will be viewed favourably by others. A survey of recreational athletes 15 years and older was distributed via social media to sports clubs and individuals. To overcome social desirability bias, the participants were questioned using the following Randomized Response Technique.

Recreational athletes participating in the survey were asked to first roll a die in private. They then responded to one of the following questions depending on the outcome of the rolled die:

- If the participant rolled a 1, then the participant responded to the question: *"Does a week have 9 days?"*
- If the participant rolled a 2, 3 or 4, then the participant responded to the question: *"In 2019 did you use over-the-counter medications to enhance your sporting performance?"*
- If the participant rolled a 5 or a 6, then the participant responded to the question: *"Does every week have 7 days?"*

a) Let *p* be the proportion of recreational athletes who have used over-the-counter medication to enhance their sporting performance in 2019. Draw a tree diagram to represent this randomised response technique. Indicate the probabilities on each branch of the tree. *[4 marks]*



b) What is the probability that a surveyed recreational athlete rolls a 2, 3 or 4, and responds "Yes". *[2 marks]*

$P(\text{Yes AND Rolls } 2, 3, \text{ or } 4)$

$= P(\text{Yes} \mid \text{Rolls } 2, 3 \text{ or } 4)\, P(\text{rolls } 2, 3 \text{ or } 4)$

$= p \times \frac{1}{2}$

c) The study found that 164 out of 420 recreational athletes surveyed responded "Yes". What is the estimated proportion of recreational athletes who used over-the-counter medication to enhance their sporting performance? *[6 marks]*

$P(\text{Yes}) = \frac{1}{6} \times 0 + \frac{1}{2} \times p + \frac{1}{3} \times 1 = \frac{1}{3} + \frac{1}{2} p$

Estimate for $P(\text{Yes}) = \frac{164}{420} \simeq 0.39$

Solve for $\hat{p}$

$\frac{1}{3} + \frac{1}{2}\hat{p} = 0.39 \quad \Rightarrow \hat{p} = 2\left(0.39 - \frac{1}{3}\right) \simeq 0.11$

Estimated proportion using over the counter medication to enhance sporting performance is 0.11.

END OF EXAMINATION

Formulas

Conditional Probability

P(A and B) = P(A | B) P(B)

If A and B are independent, then P(A and B) = P(A) P(B)

Expected Values

$$\mathrm{E}(X) = \sum_x P(X = x)x \qquad \mathrm{var}(X) = \sum_x P(X = x)\big(x - \mathrm{E}(X)\big)^2 \qquad \mathrm{sd}(X) = \sqrt{\mathrm{var}(X)}$$

$$\mathrm{E}(aX + b) = a\mathrm{E}(x) + b \qquad \mathrm{sd}(aX + b) = |a|\mathrm{sd}(X)$$

$$\mathrm{E}(X_1 + X_2) = \mathrm{E}(X_1) + \mathrm{E}(X_2) \qquad \mathrm{var}(X_1 + X_2) = \mathrm{var}(X_1) + \mathrm{var}(X_2), \text{ if independent}$$

Standardising

$$\text{If } X \sim \text{Normal}(\mu, \sigma) \text{ then } Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0,1)$$

Binomial distribution

$$\mathrm{E}(X) = np \qquad \mathrm{sd}(X) = \sqrt{np(1-p)} \qquad \hat{P} = \frac{X}{n} \qquad \mathrm{E}(\hat{P}) = p \qquad \mathrm{sd}(\hat{P}) = \sqrt{\frac{p(1-p)}{n}}$$

Tests and Confidence Intervals based on Standard Errors

$$t = \frac{\text{estimate} - \text{hypothesised}}{\text{se(estimate)}} \qquad \text{estimate} \pm t^*\text{se(estimate)}$$

$$\mathrm{se}(\bar{x}) = \frac{s}{\sqrt{n}} \qquad \mathrm{se}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \qquad \mathrm{se}(r) = \sqrt{\frac{1 - r^2}{n - 2}}$$

$$\mathrm{df} = n - 1 \qquad \mathrm{df} = \min\,(n_1 - 1, n_2 - 1) \qquad \mathrm{df} = n - 2$$

$$\mathrm{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \mathrm{se}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Pooled t-Test and Analysis of Variance

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\mathrm{DFT} = n - 1 \quad \mathrm{DFG} = k - 1 \quad \mathrm{MS} = \frac{\mathrm{SS}}{\mathrm{DF}} \quad R^2 = \frac{\mathrm{SSG}}{\mathrm{SST}} \quad s_p = \sqrt{\mathrm{MSR}} \quad F = \frac{\mathrm{MSG}}{\mathrm{MSR}}$$

$$\alpha = \frac{0.05}{k}$$

Linear Regression

$$y = b_0 + b_1 x$$

$$\mathrm{df} = n - 2$$

$$y = b_0 + b_1 x_1 + b_2 x_2 \qquad x_1 = \begin{cases} 1, & \text{if Group B} \\ 0, & \text{if Group A} \end{cases}$$

$$\mathrm{df} = n - 3$$

Odds and Logistic Regression

$$\text{odds} = \frac{p}{1-p} \qquad \mathrm{OR} = \frac{\text{odds for group B}}{\text{odds for group A}} \qquad \ln\left(\frac{p}{1-p}\right) = b_0 + b_1 x$$

Chi-Squared Tests

$$\text{expected} = \frac{(\text{row total}) \times (\text{column total})}{\text{overall total}} \qquad \chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}$$

$$\mathrm{df} = (\#\text{rows} - 1) \times (\#\text{columns} - 1)$$

R Output

You may find the following output from R to be useful in answering the questions on this examination paper.

t-distribution probabilities

```
> pt(0.98, df=35)
[1] 0.833094
> pt(0.98, df=36)
[1] 0.8331868
> pt(0.98, df=70)
[1] 0.8347695
> pt(0.98, df=71)
[1] 0.8347931
> pt(1.39, df=35)
[1] 0.9133466
> pt(1.39, df=36)
[1] 0.9134677
> pt(1.39, df=70)
[1] 0.9155333
> pt(1.39, df=71)
[1] 0.9155642
> pt(2.36, df=54)
[1] 0.9890409
> pt(2.36, df=55)
[1] 0.9890749
> pt(2.36, df=56)
[1] 0.9891077
> pt(2.36, df=57)
[1] 0.9891393
```

t-distribution quantiles

```
> qt(0.95, df=34)
[1] 1.690924
> qt(0.975, df=34)
[1] 2.032245
> qt(0.95, df=35)
[1] 1.689572
> qt(0.975, df=35)
[1] 2.030108
> qt(0.95, df=36)
[1] 1.688298
> qt(0.975, df=36)
[1] 2.028094
> qt(0.95, 54)
[1] 1.673565
> qt(0.975, 54)
[1] 2.004879
> qt(0.95, 55)
[1] 1.673034
> qt(0.975, 55)
[1] 2.004045
> qt(0.975, df=71)
[1] 1.993943
> qt(0.95, df=71)
[1] 1.6666
```

F distribution probabilities

```
> pf(6.04, 4,15)
[1] 0.9957864
> pf(6.04, 4,16)
[1] 0.9963103
> pf(6.04, 5,14)
[1] 0.996497
> pf(6.04, 5,15)
[1] 0.9970436
> pf(7.54, 4, 15)
[1] 0.9984554
> pf(7.54, 4, 16)
[1] 0.9987025
> pf(7.54, 5, 14)
[1] 0.9987321
> pf(7.54, 5, 15)
[1] 0.9989817
> pf(8.05, 4, 15)
[1] 0.9988712
> pf(8.05, 4, 16)
[1] 0.9990647
> pf(8.05, 5, 14)
[1] 0.999075
> pf(8.05, 5, 15)
[1] 0.9992689
```

Chi-squared distribution probabilities

```
> pchisq(1.48, df=1)
[1] 0.7762255
> pchisq(1.48, df=2)
[1] 0.5228861
> pchisq(1.48, df=3)
[1] 0.313106
> pchisq(1.48, df=4)
[1] 0.1698218
> pchisq(1.52, df=1)
[1] 0.7823805
> pchisq(1.52, df=2)
[1] 0.5323336
> pchisq(1.52, df=3)
[1] 0.3223379
> pchisq(1.52, df=4)
[1] 0.1769071
```